

1 Machine Unlearning — what & why

Definition: removing the **influence of training data** from a trained model — producing an unlearned model that **behaves like a model retrained** on the same data *minus* the part to forget.

Why needed: most models train on public data; retraining is very expensive; data may contain things we shouldn't have learned. We want to **edit/remove** private data, stale knowledge, copyrighted material, toxic/unsafe content, dangerous capabilities, misinformation — **without retraining from scratch**.

Exact definition depends on:

- the **ML task** (classification, language modeling...)
- the **data to unlearn** (images, articles, 'how to make napalm')
- the **algorithm** (heuristic fine-tuning vs deleting components)
- the **goal** (user privacy vs harmfulness removal)

GDPR Article 17 = Right To Be Forgotten (RTBF), 2014. Google had to remove data; removing data from ML is hard → 'data deletion' / 'machine unlearning'.

2 Two motivations

A. Access revocation (unlearn private & copyrighted data)

- Data should be 'returned', but Google **consumed** it to build the model → very hard
- May require **proof of unlearning** — hard to provide
- Analogy: Ram ate Laxman's laddu (data); can only give back something **equivalent**, not the original
- **Periodic re-training:** batch unlearning requests, apply at next retrain; policymakers can mandate economically-viable deadlines, offloading cost to owners (e.g. OpenAI)

B. Model correction & editing

- Post-training **risk-mitigation** for AI-safety concerns (toxicity, bias, stale/dangerous knowledge)
- More a **desire than a necessity** — no formal guarantees/proofs needed for usefulness
- Goal: deploy **sufficiently safe** models (e.g. chatbots)

3 Forms of Unlearning

Form	Idea	Notes
Exact unlearning	Unlearned & retrained models are distributionally identical ; retrain without the forget points	Method = SISA (Sharded, Isolated, Sliced, Aggregated)
Approx: via Differential Privacy	Unlearned & retrained models are distributionally close	If an adversary can't tell them apart, it's as if never learned
Approx: empirical, examples known	Forget set = clearly defined training examples	'training to unlearn' / fine-tuning heuristics
Approx: empirical, underspecified	Forget set = not clearly defined ('a concept', 'a fact')	= model/concept editing, model surgery, knowledge unlearning
Just ask for unlearning	Prompt the model to 'pretend not to know'	In-context / few-shot label-flipping

Assignment Q1 matching: Exact ↔ distributionally **identical**; via Differential Privacy ↔ unlearned & retrained models (distributionally close); Example unlearning ↔ forget set **clearly defined**; Concept unlearning ↔ forget set **not clearly defined**.

4 Exact Unlearning (SISA)

Benefits:

- **The algorithm is the proof** — by design, unlearned data **never contributed** to other components (split into shards/slices)
- **Interpretability by design** — we know how each data point contributed
- To unlearn a point: **retrain only the constituent model whose shard contains it** (not the whole model) → cheap

Drawbacks:

- **Sharding is hard in deep learning** → accuracy loss
- Performance **deteriorates exponentially** with the number of unlearning samples

5 Approximate: Differential Privacy & MIA

DP unlearning considerations:

- Gives **statistical guarantees**; DP is per-example, but large models don't fit that intuition
- Guarantees **fall off quickly** with more unlearning requests
- Assumes all data points equal — but some (bomb-making) get more unlearn requests than others (Disneyland)
- For LMs: distinguish unlearning **pre-training** vs **fine-tuning** data; DP fine-tuning is feasible, DP pre-training much less so

6 Concept/Knowledge Unlearning & 'Just Ask'

Concept unlearning (unknown example space): foundation models get requests to forget a 'concept/fact', which can't map to a clean example set.

- 'Biden is US president' is **dispersed** everywhere — can we unlearn all occurrences? Does it also erase family details?
- Artists / NYT can't enumerate every copy & adaptation online
- How: ask GPT-4 for **plausible alternative completions** (Harry Potter → 'Mr. Potter studies baking'); collect hate-speech examples to unlearn harmful behavior

7 Evaluating Unlearning

Metric	Question it answers
Efficiency	How fast vs full re-training? (easy to measure)
Model utility	Do we harm performance on retain data / orthogonal tasks ? (easy to measure)
Forgetting quality	How well is the forget data actually unlearned? (the key challenge)
– If forget examples are specified: expect near-chance accuracy on them (high on retain/test, low on forget) or lower likelihood of forget sequences; or run MIA and check attack success drops below a threshold – Hard because models generalize from related data; the counterfactual of 'never seeing the forget data' can be undefined (black-box nature) – Core issue: lack of datasets/benchmarks. Fix = add synthetic manipulations for a clear measurable goal (e.g. Interclass Confusion — remove synthetic confusion between two classes)	

8 Benchmarks & Baseline Methods

TOFU — fake author profiles generated by GPT-4; finetune LLM on them, then unlearn a **subset** while retaining the rest; QA pairs check knowledge before/after; checks **ripple effects**.

Corrective Machine Unlearning: mitigate impact of data with **unknown manipulations**, knowing only a subset. Most methods (even gold-standard retraining/EU) fail unless ≥80% of poisoned data is identified; only **SSD** does well at low identification.

Abbrevs: EU = exact unlearning; CF = catastrophic forgetting; SSD = Selective Synaptic Dampening; BadT = Bad Teacher; SCRUB = Scalable Remembering & Unlearning.

Membership Inference Attack (MIA):

- A classifier trained to **distinguish training data from unseen data**
- **Unlearning goal**: the MIA should classify **forget-set samples as 'unseen'**
- Methods (besides SISA): add **noise** to weights disproportionately influential for the forget set, or **gradient ascent** on the forget set

Just ask for unlearning:

- **Pretend** not to know (works best for **well-known** entities — model must know it well to fake not knowing)
- **Few-shot / in-context unlearning**: flip labels of forget examples in the prompt + add correct retain examples; model treats flipped labels as truth
- Good for factual associations; unlikely to work for toxic behaviours (output space too large)

TOFU baseline unlearning methods:

- **Gradient Ascent** — **increase** training loss on the forget set (mess up its predictions)
- **Gradient Difference** — also **decrease** loss on the retain set
- **KL** — max similarity to original model on retain, min on forget
- **DPO** — make model say **'I don't know'** on forget, correct output on retain

WMDP — unlearn **dangerous knowledge**: biosecurity, cybersecurity, chemical security; 4000+ MCQs; proposes **activation-steering** unlearning. Tied to White House EO (models shouldn't lower the barrier to chem/bio weapons). **CUT** = Contrastive Unlearning Tuning.

Assignment: Gradient Ascent = increase loss on forget set. WMDP areas = **Biosecurity, Cybersecurity, Chemical security** (NOT financial). Model Utility = does unlearning hurt retain/unrelated tasks. MIA = detect if data was in training. TOFU/WMDP are **higher-level** (knowledge) not example-level.

9 Unlearning Hardness

Difficulty scales with how **embedded** the knowledge is: infrequent occurrences (car accidents in Palo Alto) < frequent ('Biden is US president') < **fundamental facts** ('the sun rises every day'). Fundamental knowledge is **deeply embedded in the model's implicit knowledge graph** — can't be removed without introducing contradictions and harming utility. (Case study: *Who's Harry Potter?* — Microsoft/Llama2-7b-WholsHarryPotter.)

10 Representation Engineering (RepE) + LAT

Representation Engineering = modify representation vectors to control model outputs. Studies traits like honesty, power-seeking, morality; they can be **identified live at output** and **controlled**.

vs mechanistic interpretability: mechanistic is **bottom-up** (node-to-node connections); RepE looks at **larger chunks of representations / higher-level mechanisms**.

Hidden activations = intermediate vectors collected after model components run; we can inspect what info they hold and edit them to change outputs. **Crux**: LLM representations **already understand** concepts, so we find & enhance them — often better than just prompting.

LAT — Linear Artificial Tomography (3 steps):

- **Design stimulus prompts** to elicit the concept/function
- **Collect internal activations** (<1000 inputs enough) — <concept> token or last token before prediction
- **Find the concept direction** (linear model): one-shot 'M(Love)–M(Hate)'; unsupervised PCA top-1 / K-Means; supervised Contrastive PCA / class-mean-difference / linear classifier

Uses: emotion, bias (stereotypical vs non), **harmlessness** (90% accuracy classifying harmful prompts, robust even to GCG jailbreak suffixes), memorization, editing/unlearning.

Evaluate on ethical utility: Classification→correlation;

Generation→effective; Termination→necessity; Recovery→sufficiency.

11 Graph Unlearning

Unlearning on graph neural networks (**GCN** = Graph Convolutional Network; **MPNN** = Message-Passing Neural Network). Three types: **node-feature unlearning, node unlearning, edge unlearning**.

Method	Summary / disadvantage
GraphEraser	Sharding-based (node/edge/feature) — sharding is hard
GUIDE	Optimises partitioning & graph aggregation — sharding is hard
Projector	Projects weight params where forgotten features are irrelevant — linear GNN only
GraphEditor	Closed-form removal of deleted nodes + update neighbours — linear GNN only
GNNDelete	Shared weight matrices + layer-wise deletion operator — can't handle continual learning
MEGU	Mutual Evolution Graph Unlearning: adaptive high-influence neighbourhood + custom loss

Problems with graph unlearning:

- Focus on forget set may hurt retain set; models tend to **over-forget**
- Little focus on **continual learning** & deployment overhead
- Are methods truly forgetting? Are metrics tested properly? (active research area)

Unlearning for AI safety:

- Remove **hazardous knowledge** (WMDP)
- Remove **model poisons & backdoors** (trigger-based)
- Remove **manipulative behaviours** (unethical persuasion, deception)
- Remove **bias & toxicity**; remove **power-seeking** tendencies

Assignment Q12 — Edge Unlearning methods: GraphEraser, GNNDelete, MEGU. (Projector & GraphEditor are node/feature-oriented linear-GNN methods; GUIDE not counted for edges here.)